

Math 105LA Assignment 1

Fall 2018

Due Sun 10/6, 11pm

Write a MATLAB function for implementing the bisection method:

Input: f , a , b , N , tol , tol_fc

Output: c

Step 1: Set $n = 1$, $F_a = f(a)$.

Step 2: While $n \leq N$, repeat step 3 to step 6

Step 3: Set $c = (a+b)/2$, $F_c = f(c)$.

Step 4: if $|F_c| < \text{tol_fc}$

Output('f(c) is smaller than the tolerance', c)

elseif $(b-a)/2 < \text{tol}$

Output('c converged with accuracy tol', c)

end

Step 5: Set $n = n+1$

Step 6: if $F_a * F_c < 0$

set $b = c$

else

set $a = c$, $F_a = F_c$

Step 7: Output('Maximum number of iteration is reached', c)

Make sure that you use the **same order** of input below:

```
function c = bisection(f, a, b, N, tol, tol_fc)
%% type your code below

% .....
% while .....
% .....
%     if .....
%         disp('|f(c)| is smaller than the tolerance')
%         return
%     elseif .....
%         fprintf('c is converged with %0.2e accuracy\n', tol)
%     end

% .....
% end
% disp('Maximum number of iteration is reached')

end
```

Let us consider the following test example for checking the code.

For example, we will want to solve the equation $x^2 - 2 = 0$ in $[1,2]$.

Assuming the function is saved as "bisection.m". We can use the following Matlab comment to check your code:

```
f = @(x) x^2-2;
c = bisection(f,a,b,N,tol,tol_fc)
```

Input	Output	Output: c
a = 1; b = 2; N = 2; tol = 1e-3; tol_fc = 1e-8;	Maximum number of iteration is reached	c = 1.2500
a = 1; b = 2; N = 20; tol = 1e-3; tol_fc = 1e-8;	c is converged with 1.00e-03 accuracy	c = 1.4150
a = 1; b = 2; N = 20; tol = 1e-8; tol_fc = 1e-8;	Maximum number of iteration is reached	c = 1.4142
a=2*sqrt(2)-2+eps(1);b=2;N=20;tol=1e-8; tol_fc=1e-8;	f(c) is smaller than the tolerance	c = 1.4142

Another examples: f = @(x) exp(x)- 4*x^2;

Input	Output	Output: c
a = 0; b = 2; N = 2; tol = 1e-3; tol_fc = 1e-8;	Maximum number of iteration is reached	c = 0.5000
a = 0; b = 2; N = 20; tol = 1e-3; tol_fc = 1e-8;	c is converged with 1.00e-03 accuracy	c = 0.7139
a = 0; b = 2; N = 20; tol = 1e-8; tol_fc = 1e-8;	Maximum number of iteration is reached	c = 0.7148
a = 0; b = 2; N = 200; tol = 1e-8; tol_fc = 1e-8;	c is converged with 1.00e-08 accuracy	c = 0.7148

Submit a single file named by your UCInetID (e.g. wt1eung.m) that contains the function. Your code will be tested on a few problems. Your grade of this assignment is calculated by the number of problems that your code solves correctly. Make sure your function runs without errors and has **the same order of input** above. **Failure to do so will result in a zero grade.**

Some useful MATLAB Commands:

- 1) If-statement. Use the following code for "if x<0 then x=y"

```
if x<0
    y = x;
end
```

- 2) In MATLAB function, use the `return` command to terminate the execution of a function.
- 3) Use `abs(x)` to compute $|x|$.